

Learning Bounded Latent Degradation Dynamics for stable rollout and Remaining Useful Life Prediction

Arpit Goel^{a,*}

^a*TwinSim Labs,*

ARTICLE INFO

Keywords:

stable forecasting
latent representation learning
multivariate time series
autoencoders
degradation modelling
remaining useful life

ABSTRACT

Long-horizon multivariate time-series forecasting is often unstable when models are recursively rolled out in observation space. This problem is especially pronounced in degradation modelling, where sensor trajectories are noisy, regime-dependent, and only partially informative about latent health. We study whether low-dimensional bounded latent representations can act as dynamical regularizers for stable rollout. The proposed algorithm identifies operating regimes from control variables, normalizes sensor measurements within regimes, denoises the resulting trajectories, selects degradation-sensitive channels, and trains a bounded latent autoencoder with monotonicity and smoothness regularization on a primary health coordinate. We provide a theoretical analysis separating sensor-space closed-loop divergence, bounded latent rollout, and smoothness-controlled forecastability, and we extend it in two directions: an error bound for damped and windowed extrapolation, and anchored decoding, which cancels the reconstruction offset that a decoded forecast carries at its origin. Experiments on NASA C-MAPSS show that higher-dimensional latent spaces can improve reconstruction while weakening rollout stability. Ablations further show that finite-horizon decoded boundedness is not unique to the bounded latent model: an unbounded autoencoder may also remain numerically bounded under constant-velocity rollout because of small latent velocities and decoder saturation. The main empirical advantage of the proposed representation is therefore forecastability: lower latent curvature, higher forecast skill, and useful downstream remaining-useful-life prediction. On the PHM 2010 CNC milling benchmark both mechanisms transfer outside turbofan degradation: the sensor-space recursion leaves the boundedness criterion under a long free run while every bounded latent rollout stays controlled, and anchored decoding removes the short-horizon penalty that plain decoding incurs.

1. Introduction

Many multivariate time-series applications require models that remain useful when rolled forward beyond a one-step training objective. In degradation modelling and prognostics, this requirement is especially important: forecasts are used not only to interpolate observed measurements, but also to reason about future system health. A model that reconstructs historical trajectories accurately may still produce implausible or unstable behavior when recursively rolled out.

Raw sensor measurements mix several sources of variation. Measurements change with operating conditions, sensor noise, unit-specific behavior, and latent degradation. Directly forecasting in observation space can therefore lead to unstable dynamics. Linear autoregressive models may become non-contractive when their closed-loop transition operators have spectral radius greater than one (Hamilton, 1994; Lütkepohl, 2005), while high-capacity sequence models may achieve good short-horizon accuracy without guaranteeing stable free-run behavior, a difficulty long documented for recurrent architectures trained via backpropagation through time (Pascanu, Mikolov and Bengio, 2013).

This paper studies an algorithmic alternative. Instead of forecasting raw sensors directly, we learn a compact latent state intended to represent degradation after removing operating-regime effects. The latent state is bounded by construction and is regularized so that its primary coordinate evolves smoothly and approximately monotonically over the life of each unit. Forecasting is then performed by local constant-velocity rollout in latent space, followed by decoding back to the sensor representation.

*Corresponding author

✉ arpit@thebrownkid.in (A. Goel)

ORCID(s):

The key thesis is that stable long-horizon forecasting requires more than accurate reconstruction. Bounded latent geometry constrains the admissible state space, while smooth low-curvature latent dynamics make motion within that space forecastable. These are distinct mechanisms: boundedness alone is not sufficient for useful forecasting, and short-horizon accuracy does not ensure stable free-run behavior.

The main contributions are:

- We propose a regime-aware latent representation learning algorithm for multivariate degradation time series, combining condition normalization, denoising, dynamic sensor selection, and bounded latent autoencoding.
- We provide a theoretical analysis separating sensor-space closed-loop divergence, bounded latent rollout, and smoothness-controlled latent forecastability.
- We show empirically that boundedness alone is insufficient: an unbounded latent autoencoder can remain finite over practical horizons, but exhibits higher latent curvature and weaker forecast skill.
- We demonstrate that increasing latent dimension improves reconstruction but can weaken rollout stability, supporting low-dimensional latent manifolds as dynamical regularizers.
- We compare against VAR, VAR(2), GRU, LSTM, and TCN baselines, showing that closed-loop sensor-space and neural sequence forecasters can exhibit inconsistent free-run stability.
- We introduce *anchored decoding*, a reformulation of the decoded forecast that cancels the origin reconstruction offset. We prove its error obeys the same curvature bound as the underlying rollout up to the decoder Lipschitz constant, plus a reconstruction-residual term, and we give the condition on that residual under which anchoring is an improvement. Empirically it removes the short-horizon penalty that plain decoding incurs, changing horizon-one skill on CNC tool wear from -263 to -0.73 .

2. Related Work

2.1. RUL Prediction and Health Indicators

Remaining useful life (RUL) estimation is a central task in prognostics and health management, where the goal is to infer the remaining time before a system reaches end of useful life from condition-monitoring data (Si, Wang, Hu and Zhou, 2011). NASA C-MAPSS is a standard benchmark for this setting: it provides run-to-failure multivariate time series, operating settings, sensor noise, official train/test splits, and test RUL labels (NASA, 2008; Saxena and Goebel, 2008; Saxena, Goebel, Simon and Eklund, 2008). The four subsets FD001–FD004 vary in operating-condition complexity and fault-mode multiplicity.

Classical health-indicator construction emphasizes properties such as monotonicity, trendability, and prognosability (Coble and Hines, 2009). These criteria motivate representations that move consistently with degradation and can be extrapolated toward failure. More recent work explicitly builds monotonicity and other domain constraints into the learning objective for data-driven health indicators, closer in spirit to the regularization used here (Tefera, Van Baelen, Meire, Luca and Karsmakers, 2025). Our work adopts this health-indicator perspective but learns the health state as a latent coordinate rather than hand-engineering it from individual sensors.

2.2. Deep Learning for Multivariate Prognostics

Deep learning methods have been widely used for RUL prediction from multivariate sensor windows. Convolutional models learn local temporal features (Babu, Zhao and Li, 2016; Li, Ding and Sun, 2018), while recurrent architectures such as LSTMs and GRUs model sequential dependencies (Zheng, Ristovski, Farahat and Gupta, 2017; Hochreiter and Schmidhuber, 1997; Cho, van Merriënboer, Gulcehre, Bahdanau, Bougares, Schwenk and Bengio, 2014). Temporal convolutional networks provide a strong generic sequence-modelling baseline with causal dilated convolutions and residual connections (Bai, Kolter and Koltun, 2018). Attention mechanisms have also been layered onto recurrent and convolutional backbones for RUL estimation, including feature-attention LSTMs (Liu, Liu, Jia and Lin, 2021), attention-augmented LSTMs for rotating machinery (Zhang, Li and Wang, 2020), and bidirectional GRUs with temporal self-attention (Zhang, Jiang, Wu, Li and Luo, 2022). Recent C-MAPSS RUL models include CNN-LSTM-Attention (Deng and Zhou, 2024) and Transformer-TCN-Self-Attention feature fusion (Li, Luo, Liu, Tang and Miao, 2025).

Beyond turbofan data, the PHM 2010 high-speed milling benchmark (PHM Society, 2010) is a standard testbed for tool-wear prognostics from force, vibration and acoustic-emission channels, and is commonly addressed with deep sequence models trained on its labelled cutters (Zhang, Qi, Wang and He, 2023). We use it in Section 6.5 as a degradation process whose sensor dynamics differ substantially from C-MAPSS.

However, supervised RUL accuracy and stable free-run forecasting are different objectives. A sequence model can be accurate over a short horizon or under teacher forcing while becoming unstable when recursively rolled forward. In this paper, GRU, LSTM, and TCN models are treated as forecasting baselines for testing free-run stability rather than as direct replacements for the proposed latent representation.

2.3. Autoencoders and Latent Dynamics

Autoencoders are commonly used to learn compact representations of high-dimensional signals. Denoising autoencoders learn features robust to input corruption (Vincent, Larochelle, Bengio and Manzagol, 2008), and variational autoencoders introduce probabilistic latent variables via amortized stochastic inference (Kingma and Welling, 2014; Rezende, Mohamed and Wierstra, 2014). In prognostics, autoencoder representations are often used as features for downstream health or RUL prediction.

Latent dynamics methods similarly separate high-dimensional observations from lower-dimensional evolution. This perspective traces back to classical operator theory, in which a nonlinear dynamical system is represented through the linear evolution of observables in an infinite-dimensional function space (Koopman, 1931). Koopman autoencoders operationalize this idea with finite-dimensional learned embeddings in which nonlinear dynamics are approximated by linear evolution (Lusch, Kutz and Brunton, 2018; Azencot, Erichson, Lin and Mahoney, 2020). Latent Space Dynamics Identification combines autoencoder compression with learned latent dynamics for reduced-order rollout (Fries, He and Choi, 2022), with subsequent work adding Gaussian-process uncertainty over the latent dynamics (Bonneville, Choi, Ghosh and Belof, 2024). Our work shares the broad principle of forecasting in latent space, but focuses on how bounded low-dimensional geometry and latent smoothness affect free-run stability and forecastability in degradation time series.

2.4. Position of This Work

The contribution of this paper is not a new autoencoder architecture alone. Rather, we study bounded latent manifolds as dynamical regularizers. The experiments explicitly separate reconstruction, boundedness, curvature, forecast skill, and downstream RUL. This distinction is important because a model may reconstruct well but forecast poorly, and a model may remain numerically bounded over finite horizons without learning a useful degradation trajectory.

Koopman autoencoders and LaSDI share the premise that forecasting should happen in a reduced space, but they target a different data setting. Both fit a latent evolution operator to full-state snapshots of a system whose dynamics are observed consistently: LaSDI to solution snapshots of a parameterized simulation, Koopman autoencoders to repeated observations of a single autonomous system. C-MAPSS supplies neither. Each unit is a separate, right-censored trajectory with unknown initial wear, sampled under discrete operating regimes and observed only through partial noisy sensors, with no shared parameterization across units and no governing equations. These methods are therefore not baselines that can be transferred to this setting, and we do not present them as such.

3. Method

3.1. Problem Setup

Let $\mathbf{x}_{i,t} \in \mathbb{R}^p$ denote the sensor vector for unit i at cycle t , and let $\mathbf{u}_{i,t} \in \mathbb{R}^q$ denote the corresponding operating variables. The goal is to learn a low-dimensional latent state $\mathbf{h}_{i,t} \in \mathbb{R}^K$, with $K \ll p$, that captures degradation-relevant dynamics and supports stable long-horizon rollout.

The workflow is summarized in Fig. 1. It first removes nuisance variation caused by operating conditions and static sensors, then learns a compact latent state whose geometry is suitable for free-run forecasting.

3.2. Regime Identification and Normalization

Operating conditions can dominate raw sensor variation. We therefore identify discrete regimes using only the operating variables. A clustering function C assigns each observation to a regime, $r_{i,t} = C(\mathbf{u}_{i,t})$. In the implementation,

Algorithm 1: Bounded Latent Degradation Manifold Learning

Input: Trajectories $\{\mathbf{x}_{i,t}, \mathbf{u}_{i,t}\}$ and latent dimension K
Output: Bounded latent state $\mathbf{h}_{i,t}$, rollout $\hat{\mathbf{h}}_{i,t+\tau}$, and RUL features
Fit operating-regime map $r_{i,t} = C(\mathbf{u}_{i,t})$ using control variables only;
Estimate regime-specific statistics and compute normalized residuals;
Denoise residual trajectories using a rolling median;
Select dynamic sensors using degradation-trend scores;
Train bounded autoencoder $\mathbf{h}_{i,t} = E_\theta(\mathbf{y}_{i,t}) \in [0, 1]^K$, $\hat{\mathbf{y}}_{i,t} = D_\phi(\mathbf{h}_{i,t})$;
Penalize reconstruction error, negative increments of h_0 , and roughness of h_0 ;
Estimate local latent velocity and roll out $\hat{\mathbf{h}}_{i,t+\tau} = \Pi_{[0,1]^K}(\mathbf{h}_t + \tau \hat{\mathbf{v}}_t)$;
Decode $\hat{\mathbf{h}}_{i,t+\tau}$ or use $[\mathbf{h}, \hat{\mathbf{h}}]$ for RUL prediction;

C is a K-means model fit on the three operating settings. Sensors are not used to form regimes, preventing degradation-sensitive measurements from influencing the operating-condition partition.

For each regime r and sensor j , we estimate a regime-specific mean $\mu_{r,j}$ and scale $\sigma_{r,j}$. The normalized residual is

$$\tilde{x}_{i,t,j} = \frac{x_{i,t,j} - \mu_{r_{i,t},j}}{\sigma_{r_{i,t},j} + \epsilon}.$$

This removes operating-regime offsets and yields trajectories more likely to reflect degradation.

3.3. Denoising and Dynamic Sensor Selection

The normalized trajectories are denoised independently for each unit using a rolling median,

$$\bar{x}_{i,t,j} = \text{median}_{\tau \in W(t)} \tilde{x}_{i,t-\tau,j}.$$

This suppresses local spikes while preserving slowly varying degradation trends.

Not every sensor channel is degradation-informative. For each denoised sensor j , we compute

$$S_j = \frac{1}{N} \sum_i \left| \text{corr}(\bar{x}_{i,:j}, t) \right|.$$

The selected dynamic sensor set is $\mathcal{S} = \{j : S_j \geq \theta\}$, and the autoencoder input is $\mathbf{y}_{i,t} = \bar{\mathbf{x}}_{i,t,\mathcal{S}}$. Thus the autoencoder is not asked to spend capacity reconstructing static or purely operating-condition channels.

3.4. Bounded Latent Autoencoder

The encoder maps the selected denoised sensor vector to a bounded latent state:

$$\mathbf{h}_{i,t} = E_\theta(\mathbf{y}_{i,t}) \in [0, 1]^K, \quad \hat{\mathbf{y}}_{i,t} = D_\phi(\mathbf{h}_{i,t}).$$

The objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_m \mathcal{L}_{\text{mono}} + \lambda_s \mathcal{L}_{\text{smooth}}, \quad \mathcal{L}_{\text{rec}} = \sum_{i,t} \|\hat{\mathbf{y}}_{i,t} - \mathbf{y}_{i,t}\|_2^2.$$

The monotonicity and smoothness penalties are applied to the primary latent coordinate h_0 :

$$\mathcal{L}_{\text{mono}} = \sum_{i,t} \max(0, -\Delta h_{i,t,0}), \quad \mathcal{L}_{\text{smooth}} = \sum_{i,t} (\Delta h_{i,t,0})^2,$$

where $\Delta h_{i,t,0} = h_{i,t+1,0} - h_{i,t,0}$. After training, h_0 is oriented so that it increases with cycle index. The smoothness penalty follows the broader principle, used in physics-informed and constraint-regularized learning, of penalizing a differential-operator norm of the model output to encourage solutions that are physically or dynamically plausible rather than merely low-error on observed points (Raissi, Perdikaris and Karniadakis, 2019).

3.5. Latent Rollout and RUL Features

Forecasting is performed in latent space. A local velocity $\hat{\mathbf{v}}_t$ is estimated over a trailing window, and future latent states are predicted as

$$\hat{\mathbf{h}}_{t+\tau} = \Pi_{\mathcal{H}}(\mathbf{h}_t + \tau \hat{\mathbf{v}}_t), \quad \mathcal{H} = [0, 1]^K.$$

The decoded forecast is $\hat{\mathbf{y}}_{t+\tau} = D_{\phi}(\hat{\mathbf{h}}_{t+\tau})$. All bounded-rollout results use full-vector projection, matching the compact-domain assumption in Theorem 2. A legacy h_0 -only clipping rule was also tested and produced materially similar metrics.

Constant-velocity rollout is the simplest motion model that isolates the effect of latent geometry, and it is what all C-MAPSS results in Section 6 use. Where degradation does not progress at a steady rate, we also consider a learned bounded residual alternative,

$$\hat{\mathbf{h}}_{t+1} = \Pi_{\mathcal{H}}(\mathbf{h}_t + \gamma \tanh g_{\psi}([\mathbf{h}_t, \mathbf{c}_t])),$$

where g_{ψ} is a small residual MLP trained on the frozen latent trajectories with a combined one-step and multi-step free-run objective, and the step size γ caps the per-step displacement. The optional input $\mathbf{c}_t$ carries exogenous variables that are known in advance for every future step, so the head remains usable at arbitrary horizons without observing future sensors; where no such variables exist, g_{ψ} takes $\mathbf{h}_t$ alone. Because the projection $\Pi_{\mathcal{H}}$ is unchanged, Theorem 2 applies verbatim for any g_{ψ} : the learned variant changes only which motion within $\mathcal{H}$ is predicted, never whether the rollout stays in $\mathcal{H}$. This head is exercised in the step-size sweep of Appendix F. It is not of the linear-extrapolation form analysed in Proposition 1, so unlike the rules below it carries the boundedness guarantee without a curvature-based error bound, and the reported rollouts do not use it.

The reported rollouts share a single form. The velocity $\hat{\mathbf{v}}$ is estimated by ordinary least squares over a trailing window, and the forecast is

$$\hat{\mathbf{h}}_{c+\tau} = \Pi_{\mathcal{H}}(\mathbf{h}_c + \hat{\mathbf{v}} S_{\tau}),$$

where the weights S_{τ} select the rule. Taking $S_{\tau} = \tau$ gives constant velocity. Taking $S_{\tau} = (1 - \phi^{\tau})/(1 - \phi)$ damps the extrapolation geometrically toward a frozen state, in the manner of damped-trend exponential smoothing (Gardner and McKenzie, 1985); $\phi=1$ recovers constant velocity and $\phi=0$ freezes the state after a single step. A Holt damped-trend recursion, which smooths a level and a trend rather than fitting one windowed slope, is the third member of the family. The damping and smoothing constants are selected on training trajectories only. Every rule of this form obeys both Theorem 2 and Proposition 1.

For RUL prediction, the downstream regressor uses all latent coordinates and their local velocities:

$$[h_0, \dots, h_{K-1}, v_0, \dots, v_{K-1}].$$

This avoids confounding the latent-dimension study by discarding coordinates beyond h_1 .

3.6. Ablation Variants

The comparative study evaluates the full model and controlled variants: no regime normalization, no monotonicity penalty, no smoothness penalty, no monotonicity or smoothness, bounded latent autoencoder without penalties, and an unbounded latent autoencoder. These variants isolate which parts of the algorithm affect reconstruction, finite-horizon boundedness, latent curvature, forecast skill, and downstream RUL.

4. Theory

This section separates three mechanisms: sensor-space closed-loop instability, bounded latent rollout, and smoothness-controlled forecastability. The sensor-space analysis follows standard linear time-series and system-identification theory (Hamilton, 1994; Lütkepohl, 2005; Ljung, 1999), while the latent-space boundedness guarantee relies on elementary point-set topology (Munkres, 2000).

4.1. Sensor-Space Closed-Loop Instability

Consider a vector autoregression in standardized sensor space:

$$\mathbf{z}_{t+1} = \sum_{\ell=1}^p A_{\ell} \mathbf{z}_{t+1-\ell} + b.$$

With stacked state $\xi_t = [\mathbf{z}_t^{\top}, \mathbf{z}_{t-1}^{\top}, \dots]^{\top}$, the closed-loop rollout is $\xi_{t+1} = C\xi_t + \tilde{b}$, where C is the companion matrix.

Theorem 1 (Non-contractive linear rollout). *If $\rho(C) > 1$, where $\rho(C)$ is the spectral radius of the companion matrix, then the closed-loop rollout is non-contractive. For generic initial errors with nonzero projection onto the dominant eigenspace, free-running error grows geometrically at a rate governed by $\rho(C)$.*

Proof. Let e_t be the stacked forecast error. Then $e_{t+1} = Ce_t + r_t$, where r_t is the one-step residual. Iterating gives

$$e_t = C^t e_0 + \sum_{s=0}^{t-1} C^s r_{t-1-s}.$$

By Gelfand's formula, $\lim_{t \rightarrow \infty} \|C^t\|^{1/t} = \rho(C)$. If $\rho(C) > 1$, the homogeneous component grows geometrically for generic initial errors with nonzero dominant-eigenspace projection. $\square$

4.2. Bounded Latent Rollout

Let the latent forecast be $\hat{\mathbf{h}}_{c+\tau} = \Pi_{\mathcal{H}}(\mathbf{h}_c + \tau \mathbf{v}_c)$, where $\mathcal{H} \subset \mathbb{R}^K$ is compact.

Theorem 2 (Bounded latent rollout). *If $\hat{\mathbf{h}}_{c+\tau} \in \mathcal{H}$ for all $\tau \geq 0$ and the decoder D_{ϕ} is continuous on $\mathcal{H}$, then $D_{\phi}(\hat{\mathbf{h}}_{c+\tau})$ is uniformly bounded for all horizons.*

Proof. Because $\mathcal{H}$ is compact and D_{ϕ} is continuous, the image $D_{\phi}(\mathcal{H})$ is compact (Munkres, 2000). Compact subsets of finite-dimensional Euclidean space are bounded, so the decoded rollout is uniformly bounded. $\square$

Corollary 1 (Boundedness is architectural). *The boundedness guarantee depends on projection into a compact latent domain and decoder continuity. It does not depend on the monotonicity or smoothness penalty weights.*

This guarantee is a formal statement about the bounded latent model. It does not imply that every unbounded latent model must diverge over a finite horizon: finite-horizon decoded boundedness can also occur because of small extrapolation velocities or saturating nonlinearities.

4.3. Smoothness and Forecastability

Theorem 3 (Constant-velocity forecast error). *Let h_t be a scalar latent coordinate with bounded second difference $|\Delta^2 h_t| = |h_{t+1} - 2h_t + h_{t-1}| \leq \kappa$. For the constant-velocity forecast $\hat{h}_{c+\tau} = h_c + \tau(h_c - h_{c-1})$, the error satisfies*

$$|h_{c+\tau} - \hat{h}_{c+\tau}| \leq \frac{\kappa}{2} \tau(\tau + 1).$$

Proof. Let $\delta_s = h_{c+s} - h_{c+s-1}$. Then

$$h_{c+\tau} - \hat{h}_{c+\tau} = \sum_{s=1}^{\tau} (\delta_s - \delta_0).$$

Since $|\Delta^2 h_t| \leq \kappa$, we have $|\delta_s - \delta_0| \leq s\kappa$. Summing over $s = 1, \dots, \tau$ gives the bound. $\square$

Thus smoothness does not create boundedness; it reduces the curvature constant that controls constant-velocity forecast error. The same qualitative point — that a bounded second difference controls extrapolation error over a finite horizon — underlies classical forecasting practice for stochastic processes more generally (Brockwell and Davis, 2016).

Theorem 3 is stated for the textbook constant-velocity rule, in which the velocity is a single backward difference and the extrapolation weight is τ itself. The rollouts used in practice estimate the velocity over a trailing window and may damp it, so we state the bound they actually satisfy.

Proposition 1 (Damped and windowed extrapolation). *Let h_t satisfy $|\Delta^2 h_t| \leq \kappa$ and write $\delta_0 = h_c - h_{c-1}$. Consider any rollout of the form $\hat{h}_{c+\tau} = h_c + v S_\tau$, where v is an arbitrary velocity estimate and $\{S_\tau\}_{\tau \geq 1}$ are deterministic weights. Then*

$$|h_{c+\tau} - \hat{h}_{c+\tau}| \leq \frac{\kappa}{2} \tau(\tau + 1) + |\tau \delta_0 - v S_\tau|.$$

Proof. With $\delta_s = h_{c+s} - h_{c+s-1}$ we have $h_{c+\tau} - h_c = \sum_{s=1}^{\tau} \delta_s = \tau \delta_0 + \sum_{s=1}^{\tau} (\delta_s - \delta_0)$, and $|\delta_s - \delta_0| \leq s\kappa$ as in Theorem 3, so the second sum is at most $\frac{\kappa}{2} \tau(\tau + 1)$ in absolute value. Subtracting $\hat{h}_{c+\tau} - h_c = v S_\tau$ and applying the triangle inequality gives the stated bound. $\square$

Setting $v = \delta_0$ and $S_\tau = \tau$ makes the second term vanish and recovers Theorem 3. The term is therefore the price of departing from the textbook rule: a windowed velocity estimate pays $|\tau(\delta_0 - v)|$ in exchange for lower variance in v , and geometric damping in the manner of damped-trend exponential smoothing (Gardner and McKenzie, 1985) $S_\tau = (1 - \phi^\tau)/(1 - \phi)$ pays $|v| |\tau - S_\tau|$ in exchange for a rollout that cannot run away. All rollout rules reported in this paper are of this form, so all of them inherit both this error bound and the boundedness guarantee of Theorem 2.

4.4. Latent Dimension as Dynamical Regularization

Proposition 2 (Capacity-stability tradeoff). *Increasing latent dimension K weakly increases reconstruction capacity, because lower-dimensional representations can be embedded in higher-dimensional ones. At the same time, it increases the number of independent latent directions available for rollout. Therefore, larger K can improve reconstruction while weakening dynamical regularization.*

The K -aware dimension sweep tests this proposition using all latent coordinates and velocities downstream.

4.5. Anchored Decoding

The forecasts above decode the rolled-out latent directly, $D_\phi(\hat{\mathbf{h}}_{c+\tau})$, and therefore inherit a fixed reconstruction offset at the forecast origin: even a perfect latent forecast decodes to $\mathbf{y}_c$ with error $\mathbf{y}_c - D_\phi(\mathbf{h}_c)$. We remove this offset by anchoring the decoded *increment* to the last observation.

Theorem 4 (Anchored decoding forecast error). *Let $\hat{\mathbf{h}}_{c+\tau} \in \mathcal{H}$ be a latent rollout and define the anchored decoded forecast*

$$\hat{\mathbf{y}}_{c+\tau} = \mathbf{y}_c + (D_\phi(\hat{\mathbf{h}}_{c+\tau}) - D_\phi(\mathbf{h}_c)),$$

with reconstruction residual $\mathbf{r}_t = \mathbf{y}_t - D_\phi(\mathbf{h}_t)$. If D_ϕ is L -Lipschitz on $\mathcal{H}$ and each latent coordinate satisfies $|\Delta^2 h_t^{(j)}| \leq \kappa$, then

$$\|\mathbf{y}_{c+\tau} - \hat{\mathbf{y}}_{c+\tau}\| \leq L\sqrt{K} \frac{\kappa}{2} \tau(\tau + 1) + \|\mathbf{r}_{c+\tau} - \mathbf{r}_c\|.$$

If in addition $\|\mathbf{r}_t\| \leq \rho$ for all t , the last term is at most 2ρ .

Proof. Write $\mathbf{y}_t = D_\phi(\mathbf{h}_t) + \mathbf{r}_t$. The $D_\phi(\mathbf{h}_c)$ terms cancel, giving

$$\mathbf{y}_{c+\tau} - \hat{\mathbf{y}}_{c+\tau} = (D_\phi(\mathbf{h}_{c+\tau}) - D_\phi(\hat{\mathbf{h}}_{c+\tau})) + (\mathbf{r}_{c+\tau} - \mathbf{r}_c).$$

By L -Lipschitz continuity, $\|D_\phi(\mathbf{h}_{c+\tau}) - D_\phi(\hat{\mathbf{h}}_{c+\tau})\| \leq L\|\mathbf{h}_{c+\tau} - \hat{\mathbf{h}}_{c+\tau}\|$, and applying Proposition 1 coordinatewise gives $\|\mathbf{h}_{c+\tau} - \hat{\mathbf{h}}_{c+\tau}\| \leq \sqrt{K} \frac{\kappa}{2} \tau(\tau + 1)$. The triangle inequality with $\|\mathbf{r}_t\| \leq \rho$ bounds the residual term by 2ρ . $\square$

Corollary 2 (Anchoring cancels the origin offset). *Plain decoding $\hat{\mathbf{y}}_{c+\tau}^{\text{pl}} = D_\phi(\hat{\mathbf{h}}_{c+\tau})$ satisfies $\|\mathbf{y}_{c+\tau} - \hat{\mathbf{y}}_{c+\tau}^{\text{pl}}\| \leq L\sqrt{K} \frac{\kappa}{2} \tau(\tau + 1) + \|\mathbf{r}_{c+\tau}\|$. Anchoring replaces $\|\mathbf{r}_{c+\tau}\|$ by $\|\mathbf{r}_{c+\tau} - \mathbf{r}_c\|$, which vanishes at $\tau = 0$ and is small whenever the reconstruction residual is temporally persistent. The leading curvature term is unchanged, so anchoring retains the bounded-rollout guarantees of Theorems 2 and 3 while eliminating the constant reconstruction offset.*

Whether that replacement is an improvement is a property of the residual process, not a universal fact, and it is measurable before any forecast is made.

Table 1

Characteristics of the C-MAPSS subsets used in this study.

Dataset	Regimes	Faults	Train	Test	Sensors	Settings
FD001	1	1	100	100	21	3
FD002	6	1	260	259	21	3
FD003	1	2	100	100	21	3
FD004	6	2	249	248	21	3

Proposition 3 (When anchoring helps). *Suppose the reconstruction residual is weakly stationary with $\mathbb{E}\|\mathbf{r}_t\|^2 = \sigma^2$ and lag- τ autocorrelation $\rho_\tau = \mathbb{E}\langle \mathbf{r}_{c+\tau}, \mathbf{r}_c \rangle / \sigma^2$. Then*

$$\mathbb{E}\|\mathbf{r}_{c+\tau} - \mathbf{r}_c\|^2 = 2\sigma^2(1 - \rho_\tau),$$

so the anchored residual term is smaller than the plain residual term $\mathbb{E}\|\mathbf{r}_{c+\tau}\|^2 = \sigma^2$ if and only if $\rho_\tau > \frac{1}{2}$.

The consequence is practical. A model whose reconstruction error is large but temporally persistent, so that it acts as a slowly varying bias, gains from anchoring in proportion to that persistence. A model whose reconstruction error behaves like white noise loses: at $\rho_\tau = 0$ the anchored residual term is twice the plain one in mean square, a factor $\sqrt{2}$ in norm. Since ρ_τ is estimated directly from the residuals of a trained autoencoder, without retraining or rolling anything out, the decision between plain and anchored decoding can be made in advance for a given dataset and horizon rather than discovered by trial.

5. Experimental Setup

5.1. Datasets and Protocol

We evaluate the algorithm on the NASA C-MAPSS turbofan degradation benchmark (NASA, 2008; Saxena and Goebel, 2008; Saxena et al., 2008). The benchmark consists of four run-to-failure subsets, FD001–FD004. Each engine trajectory starts from an unknown initial wear state, operates normally at the beginning, develops a fault, and is observed until failure in the training split. The test split is truncated prior to failure and is accompanied by official RUL labels.

The subsets form a useful complexity ladder. FD001 and FD003 have one operating regime, while FD002 and FD004 have six regimes. FD003 and FD004 have two fault modes. All subsets contain 21 sensor channels and three operating settings.

5.2. Models and Baselines

We compare the proposed bounded latent manifold against sensor-space VAR and VAR(2), GRU, LSTM, and TCN forecasting models, unbounded and partially regularized autoencoder variants, higher-dimensional latent variants ($K = 1, \dots, 6$), and specialist versus pooled generalized models. The baseline comparison tests free-run stability, not merely one-step prediction accuracy. VAR models are closed-loop sensor-space recursions. Neural sequence baselines are rolled out recursively. The manifold model is rolled out in latent space and decoded.

5.3. Metrics

We evaluate reconstruction, stability, forecastability, and downstream RUL. Reconstruction is measured by R^2 . Free-run stability is measured using rollout growth ratios and a boundedness indicator. Forecastability is measured by NRMSE at multiple horizons, skill relative to persistence, and latent curvature κ . RUL performance is measured using RMSE, MAE, R^2 , and the NASA asymmetric score.

Final algorithm results use the production training budget. Comparative ablation sweeps use a reduced epoch budget for tractability; their ordering and qualitative conclusions match the production run. The hyperparameter sensitivity study in Appendix F uses the full production budget of 4000 autoencoder epochs over three seeds. All retraining scripts use fixed seeds, with seed robustness evaluated over seeds $\{0, 1, 2, 3, 4\}$.

Table 2

Final model performance across C-MAPSS.

Data	Reg.	ρ_2	Rec. R^2	ρ_{VAR}	VAR Gr.	Man. Norm	RUL RMSE	RUL R^2
FD001	1	.965	.930	1.020	760	5.99	14.53	.878
FD002	6	.795	.865	1.016	468	2.29	27.02	.748
FD003	1	.851	.960	1.018	57.6	4.33	16.31	.845
FD004	6	.849	.930	1.015	25.0	3.58	27.58	.744

Table 3

Free-run growth by model. Values are final/initial norm ratios.

Data	Ours	VAR	VAR(2)	LSTM	GRU	TCN
FD001	1.5	760	748	2.9	2.8	3.2
FD002	1.5	468	476	4.7	13.0	1.27×10^6
FD003	1.3	58	57	0.9	1.5	2.8
FD004	0.8	25	27	3.4	1.9	1.4

6. Results

The results are organized around the algorithmic mechanism. We first establish the learned latent geometry, then evaluate final performance, free-run stability, dimension, ablations, robustness, transfer, and RUL context.

6.1. Latent Manifold Quality

A stable latent rollout is only meaningful if the representation organizes degradation trajectories coherently. Fig. 2 shows the intrinsic-dimensionality diagnostics and learned two-dimensional latent manifolds for FD001–FD004.

The first two principal components explain 96.5%, 79.5%, 85.1%, and 84.9% of variance on FD001–FD004, respectively. FD001 has the cleanest two-dimensional structure, while FD002 and FD004 are more complex because of multiple operating regimes. Nevertheless, the learned latent coordinates organize trajectories by life fraction across all four subsets.

6.2. Main Cross-Dataset Performance

Table 2 summarizes the final bounded $K = 2$ model.

Across all datasets, the model reconstructs selected degradation-sensitive channels with mean reconstruction R^2 between 0.865 and 0.960. The fitted sensor-space VAR models are non-contractive on every dataset, with $\rho_{\text{VAR}} > 1$, and VAR free-run growth ranges from about $25\times$ to $760\times$. The bounded latent manifold remains controlled and supports RUL prediction with RMSE from 14.53 to 27.58 cycles.

6.3. Free-Run Stability Against Baselines

The baseline study asks whether stable rollout is specific to the proposed latent geometry or also appears in standard sensor-space and neural forecasters.

Fig. 3 summarizes the same comparison visually. The manifold remains bounded on all four datasets. VAR and VAR(2) diverge on every dataset, matching Theorem 1. Neural baselines are more flexible but not uniformly stable: TCN diverges catastrophically on FD002, and GRU grows by about $13\times$ on FD002.

At very short horizons, VAR can have competitive NRMSE because it is optimized as a local linear predictor. The claim is therefore not strict one-step superiority; it is stable long-horizon free-run behavior.

6.4. Dataset Requirements for Cross-Domain Transfer

A candidate dataset for this algorithm must supply four things: operating conditions or settings that are recorded separately from the measurements, so that regimes can be identified and normalized out; noisy sensor channels, so that denoising and trend-based selection have something to act on; a degradation process, so that the health coordinate has a direction to track; and enough temporal structure for the signal to be forecastable at the horizons of interest. Official remaining-useful-life labels are *not* required. RUL is a downstream use of the latent state, not a precondition for learning it, and a dataset that satisfies the first four properties can exercise reconstruction, boundedness, curvature, and forecast skill without them.

Table 4

CNC milling free-run growth. Values are final/initial decoded norm ratios at two probe horizons: the data horizon (188 steps, the last observed cut) and the end-of-life horizon (400 steps, matching the C-MAPSS protocol of Table 3). Rows marked **Ours** are the proposed bounded latent model under the rollout rules of Section 3.

Model		@188	Bnd.	@400	Bnd.
persistence	ref.	1.00	yes	1.00	yes
sensor VAR	ref.	2.69	yes	9104	no
constant velocity	Ours	2.89	yes	1.62	yes
damped CV	Ours	1.20	yes	0.98	yes
Holt damped trend	Ours	1.52	yes	1.96	yes
windowed trend	Ours	1.39	yes	1.00	yes
anchored damped CV	Ours	1.13	yes	0.98	yes
anchored Holt	Ours	1.43	yes	2.14	yes

Table 5

CNC milling forecast skill relative to persistence. Positive values beat persistence. Rows marked **Ours** are the proposed bounded latent model; anchored rows use the decoding of Section 4.5.

Model		@1	@10	@25	@50
persistence	ref.	0.00	0.00	0.00	0.00
sensor VAR	ref.	-9.51	-3.71	-2.63	-2.86
constant velocity	Ours	-263.2	-9.16	-5.06	-0.87
damped CV	Ours	-263.2	-6.38	-2.43	-0.35
Holt damped trend	Ours	-267.4	-7.07	-3.47	-0.35
windowed trend	Ours	-263.4	-7.42	-3.81	-0.74
anchored Holt	Ours	-1.35	-1.74	-1.78	-0.12
anchored damped CV	Ours	-0.73	-1.04	-0.86	+0.05

6.5. Cross-Domain Validation: CNC Tool Wear

The results above are confined to turbofan degradation. To test whether the two mechanisms of Section 4 survive a change of domain, we apply the identical pipeline to the PHM 2010 CNC milling benchmark (PHM Society, 2010): six cutting tools run to 315 cuts each, with thirty-five time-domain force, vibration and acoustic-emission features per cut, four tools used for training and two held out. The latent dimension is $K=4$ by the intrinsic-dimensionality elbow and reconstruction R^2 is 0.730; the fitted sensor-space companion matrix has spectral radius $\rho = 1.030$, so Theorem 1 predicts geometric divergence of the closed-loop sensor recursion.

Table 4 reproduces the central stability result outside C-MAPSS, and it also shows why the probe horizon matters. Stopped at the last observed cut, every model including the VAR appears bounded, and the ordering is uninformative: plain constant-velocity rollout (2.89) looks slightly worse than the VAR (2.69). Continued to the end-of-life horizon used throughout Section 6, the ordering inverts decisively. The sensor-space VAR grows by a factor of 9 104 and leaves the boundedness criterion, exactly as Theorem 1 requires of a non-contractive companion matrix, while every bounded latent rollout remains controlled and two of them contract. The distinction is not visible at short probe horizons because a spectral radius of 1.030 needs several hundred steps to express itself; it is a property of the operator, not of the window in which it is observed.

Table 5 is the second mechanism, and it is the clearest demonstration of Corollary 2 in the paper. Every plain-decoded rollout is catastrophic at horizon one, with skill near -263 ; in NRMSE the error is 0.595 against persistence at 0.0366, a factor of sixteen. That penalty is not caused by the latent dynamics, which at horizon one have barely moved. It is the constant origin offset $\mathbf{r}_c = \mathbf{y}_c - D_\phi(\mathbf{h}_c)$ that plain decoding carries at every horizon. Anchoring cancels it, and the same rollout, decoded differently, moves from -263.2 to -0.73 : an NRMSE of 0.0482, within a third of persistence rather than sixteen times worse. Anchored damped constant velocity is also the only rule with positive skill at the longest horizon.

Two features of this dataset explain why both mechanisms are visible on it at once. The fitted damping is $\phi^* = 0.9$, close to undamped extrapolation, so the latent rollout is genuinely doing work rather than contributing a single frozen step; and the reconstruction residual is large enough that the offset term dominates the curvature term at short horizons, which is the regime in which Proposition 3 predicts anchoring to pay. The two mechanisms of Section 4 are therefore

Table 6

Mechanism ablation on FD002.

Variant	Rec. R^2	κ	Skill	RUL RMSE
No regime norm.	.999	—	-0.13	40.5
Unbounded AE	—	.0346	.276	—
Full model	.865	.0019	.676	27.02

Table 7

Open-loop versus closed-loop rollout diagnostic.

Model	Space	Rollout	Behavior
VAR/VAR(2)	sensor	closed-loop	diverges, $\rho > 1$
Unbounded AE	latent	open-loop CV	finite over horizon
Bounded AE	latent	projected CV	formally bounded

both exercised on a single external dataset: bounded latent geometry controls the admissible state space over long free runs, and the decoding choice controls the error inside it.

6.6. Latent Dimension: Capacity Versus Stability

The K -aware dimension sweep tests whether the two-dimensional latent state is a bottleneck or a dynamical regularizer. For each K , all latent coordinates and velocities are used downstream, so higher-dimensional models are not evaluated only through (h_0, h_1) .

The results support Proposition 2. Reconstruction generally improves as K increases, but stability and RUL do not improve monotonically. On FD001, $K = 2$ achieves horizon-50 NRMSE about 0.465, while $K = 4$ and $K = 6$ rise to about 0.561 and 0.555. On FD002, larger latent spaces become unbounded for $K \geq 3$ under the stability criterion, even as reconstruction improves. Thus the low-dimensional latent state acts as a dynamical regularizer.

6.7. Ablation: Boundedness Versus Forecastability

The ablation study separates reconstruction, finite-horizon boundedness, forecastability, and RUL utility.

The most important ablation result is negative but clarifying. Under local constant-velocity latent rollout, the unbounded autoencoder remains finite over the evaluated horizon because the rollout is open-loop with small velocities and the tanh decoder saturates outside the training range. The empirical advantage of the proposed model is therefore not that every unbounded latent model diverges. The advantage is forecastability.

On FD002, the unbounded autoencoder has curvature $\kappa = 0.0346$ and forecast skill 0.276. The full model reduces curvature to 0.0019 and improves skill to 0.676. The no-regime-normalization variant reaches reconstruction $R^2 = 0.999$ on FD002 but collapses to forecast skill -0.13 and RUL RMSE 40.5. This directly shows that reconstruction is not prognosis.

6.8. Open-Loop Versus Closed-Loop Rollout

This diagnostic clarifies why finite-horizon decoded boundedness can occur even for the unbounded autoencoder. VAR and VAR(2) are recursive closed-loop sensor-space models. The dramatic divergence in the baseline study is associated with observation-space recursion and non-contractive transition matrices, not with every unbounded latent parameterization.

6.9. Robustness and Transfer

The main conclusions persist across five random seeds. The full bounded model remains bounded in 100% of seed runs. FD001 RUL is 14.39 ± 0.45 , and FD002 RUL is 27.40 ± 0.13 . On FD002, the unbounded autoencoder has forecast skill -0.73 ± 0.82 , while the full model achieves 0.52 ± 0.13 .

The generalized model preserves bounded rollout on three of four datasets and achieves near-identical FD002 RUL to the specialist model. This suggests that the learned geometry is partially reusable across related degradation settings. Section 6.5 tests the same pipeline outside the turbofan domain, on CNC tool wear.

Table 8

Compact robustness and transfer summary.

Study	Dataset	Key metric	Value
Seed robustness	FD001	Full RUL RMSE	14.39 ± 0.45
Seed robustness	FD002	Full RUL RMSE	27.40 ± 0.13
Seed robustness	FD002	Unbounded AE skill	-0.73 ± 0.82
Seed robustness	FD002	Full skill	0.52 ± 0.13
Transfer	FD002	Specialist RUL RMSE	26.87
Transfer	FD002	Generalized RUL RMSE	26.92

Table 9

C-MAPSS RUL context. Values are RMSE in cycles.

Method	FD001	FD002	FD003	FD004
Ours	14.53	27.02	16.31	27.58
AFICv (Alomari, Andó and Baptista, 2023)	11.80	23.00	14.60	22.30
CNN-LSTM-Att. (Deng and Zhou, 2024)	15.98	14.45	13.91	16.64
TTSNet (Li et al., 2025)	11.02	13.25	11.06	18.26

6.10. Downstream RUL Utility and Literature Context

Although the algorithm is not optimized solely as an RUL leaderboard model, the learned latent state remains useful for downstream prognosis. The final model reduces mean-baseline RMSE from 43.07–54.90 cycles to 14.53–27.58 cycles across FD001–FD004.

Table 9 places the RUL numbers in context. External numbers are reported by the cited papers and are not necessarily directly comparable because preprocessing, RUL capping, validation choices, and target definitions differ. The table is included only to show that the latent state has useful downstream RUL signal; the central contribution remains stable long-horizon rollout.

7. Discussion

7.1. Boundedness Is Not Enough

The ablation study shows that finite-horizon boundedness is not the complete explanation for stable prognostic performance. Under constant-velocity latent rollout, an unbounded autoencoder can remain numerically bounded over practical horizons because the latent velocities are small and the tanh decoder saturates. Thus, the empirical advantage of the proposed model is not merely that its decoded trajectories remain finite.

The more important distinction is between boundedness and forecastability. The bounded latent model provides a formal horizon-independent guarantee when the full latent vector is projected into a compact domain. Smoothness and monotonicity then reduce latent curvature, making constant-velocity forecasting more accurate. Boundedness controls the admissible state space; smoothness controls the predictability of motion within that state space.

7.2. Reconstruction Is Not Prognosis

Several experiments show that reconstruction accuracy can be misleading. The no-regime-normalization ablation on FD002 achieves reconstruction $R^2 = 0.999$ but has negative forecast skill and poor RUL. Similarly, increasing latent dimension improves reconstruction but can weaken rollout stability. This supports the view that degradation representations should be evaluated by forecastability and downstream prognostic utility, not reconstruction alone.

7.3. Algorithmic Scope

The proposed method is an algorithm for learning stable latent degradation states from multivariate time series with operating variables. C-MAPSS is used as a testbed because it provides controlled variation in operating regimes and fault modes. The algorithmic claim is not that the method is the best possible RUL regressor on C-MAPSS. The claim is that bounded low-dimensional latent geometry and low-curvature latent motion improve the reliability of long-horizon rollout, while preserving useful downstream RUL information.

7.4. Limitations

The evaluation covers turbofan degradation (FD001–FD004) and CNC tool wear (Section 6.5). Downstream RUL is reported only on C-MAPSS, since the milling benchmark carries no official remaining-useful-life labels, so the cross-domain evidence covers stability and forecastability but not RUL accuracy. Cross-domain evidence rests on a single external benchmark.

Anchored decoding is not universally preferable, and Proposition 3 states the condition under which it helps. Where the reconstruction residual is weakly correlated across the forecast horizon the anchored bound is the looser of the two, so the choice between plain and anchored decoding is a measurable property of a dataset rather than a universal improvement.

The latent rollout model in the main results is also deliberately simple. Constant-velocity extrapolation highlights the role of latent geometry, and the learned bounded residual head of Section 3 is exercised only in the appendices; richer latent dynamics may improve accuracy further when degradation is highly nonlinear or abrupt.

8. Conclusion

We presented a bounded latent manifold algorithm for stable multivariate time-series forecasting in degradation settings. The method removes operating-regime effects, denoises and filters degradation-sensitive sensors, and learns a compact latent state with monotonicity and smoothness regularization. Theoretical analysis separates sensor-space closed-loop divergence, bounded latent rollout, and smoothness-controlled forecastability. Experiments show that low-dimensional bounded manifolds act as dynamical regularizers: larger latent dimensions can reconstruct better but roll out worse, and boundedness alone is not sufficient without smooth latent motion. The resulting representation provides stable long-horizon behavior and useful downstream RUL prediction.

Acknowledgements

This section is intentionally left blank for submission and may be completed in the camera-ready version if appropriate.

References

- Alomari, Y., Andó, M., Baptista, M.L., 2023. Advancing aircraft engine rul predictions: An interpretable integrated approach of feature engineering and aggregated feature importance. *Scientific Reports* 13, 13466. doi:10.1038/s41598-023-40315-1.
- Azencot, O., Erichson, N.B., Lin, V., Mahoney, M.W., 2020. Forecasting sequential data using consistent koopman autoencoders, in: *Proceedings of the 37th International Conference on Machine Learning*, pp. 475–485.
- Babu, G.S., Zhao, P., Li, X.L., 2016. Deep convolutional neural network based regression approach for estimation of remaining useful life, in: *Database Systems for Advanced Applications*, Springer. pp. 214–228. doi:10.1007/978-3-319-32025-0_14.
- Bai, S., Kolter, J.Z., Koltun, V., 2018. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint arXiv:1803.01271*.
- Bonneville, C., Choi, Y., Ghosh, D., Belof, J.L., 2024. GPLaSDI: Gaussian process-based interpretable latent space dynamics identification through deep autoencoder. *Computer Methods in Applied Mechanics and Engineering* 418, 116535. doi:10.1016/j.cma.2023.116535.
- Brockwell, P.J., Davis, R.A., 2016. *Introduction to Time Series and Forecasting*. 3rd ed., Springer. doi:10.1007/978-3-319-29854-2.
- Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., Bengio, Y., 2014. Learning phrase representations using rnn encoder-decoder for statistical machine translation, in: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, pp. 1724–1734. doi:10.3115/v1/D14-1179.
- Coble, J., Hines, J.W., 2009. Identifying optimal prognostic parameters from data: A genetic algorithms approach, in: *Annual Conference of the Prognostics and Health Management Society*.
- Deng, S., Zhou, J., 2024. Prediction of remaining useful life of aero-engines based on cnn-lstm-attention. *International Journal of Computational Intelligence Systems* 17. doi:10.1007/s44196-024-00639-w.
- Fries, W.D., He, X., Choi, Y., 2022. LaSDI: Parametric latent space dynamics identification. *Computer Methods in Applied Mechanics and Engineering* 399, 115436. doi:10.1016/j.cma.2022.115436.
- Gardner, Jr., E.S., McKenzie, E., 1985. Forecasting trends in time series. *Management Science* 31, 1237–1246. doi:10.1287/mnsc.31.10.1237.
- Hamilton, J.D., 1994. *Time Series Analysis*. Princeton University Press.
- Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. *Neural Computation* 9, 1735–1780. doi:10.1162/neco.1997.9.8.1735.
- Kingma, D.P., Welling, M., 2014. Auto-encoding variational bayes, in: *International Conference on Learning Representations*.
- Koopman, B.O., 1931. Hamiltonian systems and transformation in hilbert space. *Proceedings of the National Academy of Sciences of the United States of America* 17, 315–318. doi:10.1073/pnas.17.5.315.
- Li, X., Ding, Q., Sun, J.Q., 2018. Remaining useful life estimation in prognostics using deep convolution neural networks. *Reliability Engineering & System Safety* 172, 1–11. doi:10.1016/j.res.s.2017.11.021.

- Li, Z., Luo, S., Liu, H., Tang, C., Miao, J., 2025. TTSNet: Transformer-temporal convolutional network-self-attention with feature fusion for prediction of remaining useful life of aircraft engines. *Sensors* 25, 432. doi:10.3390/s25020432.
- Liu, H., Liu, Z., Jia, W., Lin, X., 2021. Remaining useful life prediction using a novel feature-attention-based end-to-end approach. *IEEE Transactions on Industrial Informatics* 17, 1197–1207. doi:10.1109/TII.2020.2991804.
- Ljung, L., 1999. *System Identification: Theory for the User*. 2nd ed., Prentice Hall.
- Lusch, B., Kutz, J.N., Brunton, S.L., 2018. Deep learning for universal linear embeddings of nonlinear dynamics. *Nature Communications* 9. doi:10.1038/s41467-018-07210-0.
- Lütkepohl, H., 2005. *New Introduction to Multiple Time Series Analysis*. Springer. doi:10.1007/978-3-540-27752-1.
- Munkres, J.R., 2000. *Topology*. 2nd ed., Prentice Hall.
- NASA, 2008. CMAPSS Jet Engine Simulated Data. NASA Open Data Portal. Accessed 2026.
- Pascanu, R., Mikolov, T., Bengio, Y., 2013. On the difficulty of training recurrent neural networks. *Proceedings of the 30th International Conference on Machine Learning (PMLR)* 28, 1310–1318.
- PHM Society, 2010. 2010 PHM Society Conference Data Challenge. High-speed CNC milling tool-wear run-to-failure data, <https://www.phmsociety.org/competition/phm/10>. Accessed 2026.
- Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics* 378, 686–707. doi:10.1016/j.jcp.2018.10.045.
- Rezende, D.J., Mohamed, S., Wierstra, D., 2014. Stochastic backpropagation and approximate inference in deep generative models, in: *Proceedings of the 31st International Conference on Machine Learning (PMLR)*, pp. 1278–1286.
- Saxena, A., Goebel, K., 2008. Turbofan engine degradation simulation data set. NASA Ames Prognostics Data Repository. NASA Prognostics Center of Excellence.
- Saxena, A., Goebel, K., Simon, D., Eklund, N., 2008. Damage propagation modeling for aircraft engine run-to-failure simulation, in: *Proceedings of the 1st International Conference on Prognostics and Health Management*, Denver, CO, USA.
- Si, X.S., Wang, W., Hu, C.H., Zhou, D.H., 2011. Remaining useful life estimation – a review on the statistical data driven approaches. *European Journal of Operational Research* 213, 1–14. doi:10.1016/j.ejor.2010.11.018.
- Tefera, Y., Van Baelen, Q., Meire, M., Luca, S., Karsmakers, P., 2025. Constraint-guided learning of data-driven health indicator models: An application on bearings. *International Journal of Prognostics and Health Management* 16. doi:10.36001/ijphm.2025.v16i2.4268.
- Vincent, P., Larochelle, H., Bengio, Y., Manzagol, P.A., 2008. Extracting and composing robust features with denoising autoencoders, in: *Proceedings of the 25th International Conference on Machine Learning*, pp. 1096–1103. doi:10.1145/1390156.1390294.
- Zhang, H., Li, X., Wang, Y., 2020. Attention-based LSTM network for rotatory machine remaining useful life prediction. *IEEE Access* 8, 132188–132199. doi:10.1109/ACCESS.2020.3010068.
- Zhang, J., Jiang, Y., Wu, S., Li, X., Luo, H., 2022. Prediction of remaining useful life based on bidirectional gated recurrent unit with temporal self-attention mechanism. *Reliability Engineering & System Safety* 221, 108297. doi:10.1016/j.ress.2021.108297.
- Zhang, Y., Qi, X., Wang, T., He, Y., 2023. Tool wear condition monitoring method based on deep learning with force signals. *Sensors* 23, 4595. doi:10.3390/s23104595.
- Zheng, S., Ristovski, K., Farahat, A., Gupta, C., 2017. Long short-term memory network for remaining useful life estimation, in: *2017 IEEE International Conference on Prognostics and Health Management*, pp. 88–95. doi:10.1109/ICPHM.2017.7998311.

A. Full Experimental Tables

The main text reports compact summaries to avoid page-wide tables. The full machine-readable CSV files and LaTeX tables are included in the artifact bundle under `paper_artifacts/acml_results_bundle/`. They include the full baseline table, full K -sweep, full ablation table, seed robustness table, specialist-versus-generalized table, and implementation diagnostic table.

B. Additional Rollout Diagnostics

Fig. 7 shows the VAR eigenvalue spectra used to validate the non-contractive sensor-space rollout analysis. Fig. 8 shows the corresponding free-run divergence traces.

C. Forecastability Diagnostics

D. RUL Diagnostics

E. Additional Latent Geometry

F. Hyperparameter Sensitivity Analysis

This appendix reports a full-budget sensitivity analysis for the four core hyperparameters of the algorithm: the monotonicity penalty λ_m , the smoothness penalty λ_s , the latent-rollout step size γ (the residual dynamics coefficient α),

Table 10

Sensitivity to the monotonicity penalty λ_m (with $\lambda_s=2$). Full budget, mean over three seeds. Skill@20 is constant-velocity forecast skill of the health coordinate relative to persistence; κ is latent curvature (lower is smoother).

Data	λ_m	Recon R^2	κ	Skill@20	RUL RMSE
FD001	0	0.969	2.9×10^{-3}	+0.735	14.94
FD001	1	0.969	1.5×10^{-3}	+0.749	14.67
FD001	5	0.969	6.2×10^{-4}	+0.751	14.47
FD001	10	0.968	3.5×10^{-4}	+0.769	14.62
FD002	0	0.919	5.8×10^{-3}	-0.092	28.04
FD002	1	0.918	2.4×10^{-3}	+0.286	27.63
FD002	5	0.914	1.5×10^{-3}	+0.652	27.08
FD002	10	0.911	8.5×10^{-4}	+0.550	27.28

Table 11

Sensitivity to the smoothness penalty λ_s (with $\lambda_m=5$). Full budget, mean over three seeds.

Data	λ_s	Recon R^2	κ	Skill@20	RUL RMSE
FD001	0	0.968	6.2×10^{-4}	+0.752	14.16
FD001	0.5	0.969	6.1×10^{-4}	+0.752	14.22
FD001	2	0.969	6.2×10^{-4}	+0.751	14.47
FD001	5	0.969	6.4×10^{-4}	+0.754	14.22
FD002	0	0.915	1.6×10^{-3}	+0.648	27.10
FD002	0.5	0.906	1.5×10^{-3}	+0.649	27.16
FD002	2	0.914	1.5×10^{-3}	+0.652	27.08
FD002	5	0.915	1.5×10^{-3}	+0.654	27.02

and the number of normalization regimes. Each parameter is swept one at a time around the production defaults ($\lambda_m=5$, $\lambda_s=2$, $\gamma=0.05$, regimes selected automatically), with every configuration trained at the full production budget of 4000 autoencoder epochs and averaged over three seeds. Representative single-regime (FD001) and multi-regime (FD002) subsets are used, so the trends below are obtained under the same schedule as the final model. Fig. 19 summarizes all four sweeps.

The monotonicity penalty is the most consequential of the two autoencoder penalties. On the single-regime FD001 subset reconstruction is essentially flat across λ_m while curvature falls monotonically, so larger values trade nothing measurable for a smoother health coordinate. On the multi-regime FD002 subset the effect is sharper: with $\lambda_m=0$ the health coordinate is not constrained to progress with life and forecast skill collapses (mean -0.09, with high seed variance), whereas the production value $\lambda_m=5$ recovers stable positive skill (+0.65). Very large penalties ($\lambda_m=10$) begin to erode reconstruction. The default sits at the knee of this tradeoff.

The smoothness penalty is a low-sensitivity knob. Across the full range on both subsets, reconstruction, curvature, forecast skill, and RUL are all essentially unchanged. This indicates that once the monotonicity constraint fixes the orientation of the health coordinate, the additional roughness penalty is a mild regularizer rather than a delicate tuning parameter.

The step size γ separates a stable band from an unstable one rather than identifying a sharp optimum. Every value up to 0.10 yields positive skill at all three horizons on both subsets, while the largest value tested, $\gamma=0.25$, destabilizes the recursive rollout and produces strongly negative short-horizon skill. Within the stable band the settings are not separated by three seeds: on FD001 the spread at $\gamma=0.05$ (skill@10 0.24 ± 0.38) overlaps both $\gamma=0.02$ (0.41 ± 0.09) and $\gamma=0.10$ (0.43 ± 0.12), and on FD002 the same three settings are indistinguishable. The step size therefore needs to be kept below the stability threshold and does not otherwise require tuning.

The regime count is the most safety-critical hyperparameter, and it is also the clearest illustration of the paper's central distinction between reconstruction and prognosis. On the single-regime FD001 subset the number of regimes is irrelevant: forcing up to six clusters on data with one operating condition leaves every metric unchanged. On the six-condition FD002 subset the behaviour is the opposite and is initially counter-intuitive. Using too few regimes gives *higher* reconstruction ($R^2=0.986$ at one regime) but negative forecast skill and RUL error above 44 cycles, because the autoencoder spends its capacity reproducing the operating-condition staircase rather than the degradation signal. Only when the regime count matches the six physical operating conditions does forecast skill become positive (+0.65) and RUL error drop to 27 cycles, despite a lower reconstruction R^2 . High reconstruction with too few regimes is therefore actively misleading.

Table 12

Sensitivity to the latent step size γ (α). Full budget, mean over three seeds. Entries are learned-dynamics rollout skill of the health coordinate relative to persistence at horizons 10, 25, and 50.

Data	γ	Skill@10	Skill@25	Skill@50
FD001	0.02	+0.41	+0.59	+0.63
FD001	0.05	+0.24	+0.40	+0.47
FD001	0.10	+0.43	+0.63	+0.64
FD001	0.25	-8.99	-1.28	+0.06
FD002	0.02	+0.29	+0.62	+0.61
FD002	0.05	+0.33	+0.67	+0.68
FD002	0.10	+0.33	+0.66	+0.68
FD002	0.25	-5.39	-0.53	+0.19

Table 13

Sensitivity to the number of normalization regimes. Full budget, mean over three seeds. FD001 is physically single-regime; FD002 has six operating conditions. "Bnd." is the fraction of the three seeds whose decoded free-run growth stays under the boundedness threshold.

Data	Regimes	Recon R^2	Skill@20	RUL RMSE	Growth	Bnd.
FD001	1	0.969	+0.751	14.47	1.16	3/3
FD001	2	0.967	+0.754	14.21	1.16	3/3
FD001	3	0.969	+0.751	14.26	1.25	3/3
FD001	6	0.969	+0.754	14.14	1.43	3/3
FD002	1	0.986	-0.294	44.64	1.83	3/3
FD002	2	0.962	-13.67	40.27	3.39	2/3
FD002	3	0.901	-0.858	32.00	12.69	0/3
FD002	6	0.914	+0.652	27.08	2.04	3/3

The growth column shows the same effect on the stability side. At three regimes on FD002 the decoded free-run growth reaches 12.7 and falls outside the boundedness flag in all three seeds, and at two regimes in one seed of three. This does not violate Theorem 2, which concerns the image $D_\phi(\mathcal{H})$ of the compact latent box and holds whatever the regime count; the flag is the stricter empirical criterion used throughout, namely whether the decoded trajectory norm grows by less than a factor of five over a 400-step free run. A mismatched regime count leaves residual operating-condition structure in the normalized signal, the health coordinate is driven against the faces of the box, and the decoded trajectory traverses a large part of a set that is nonetheless compact. The architectural guarantee survives a badly chosen regime count; the useful behaviour does not.

G. Implementation Diagnostics

The theory-matched full-box latent projection and the legacy h_0 -only projection are compared in the implementation diagnostic artifact `theory_impl_rollout_compare`, which shows materially similar rollout metrics for the two projection rules.

Learning Bounded Latent Degradation Dynamics

Proposed pipeline: regime-aware preprocessing → bounded latent representation → stable rollout & RUL



Figure 1: Regime-aware bounded latent manifold learning.

D2_manifold

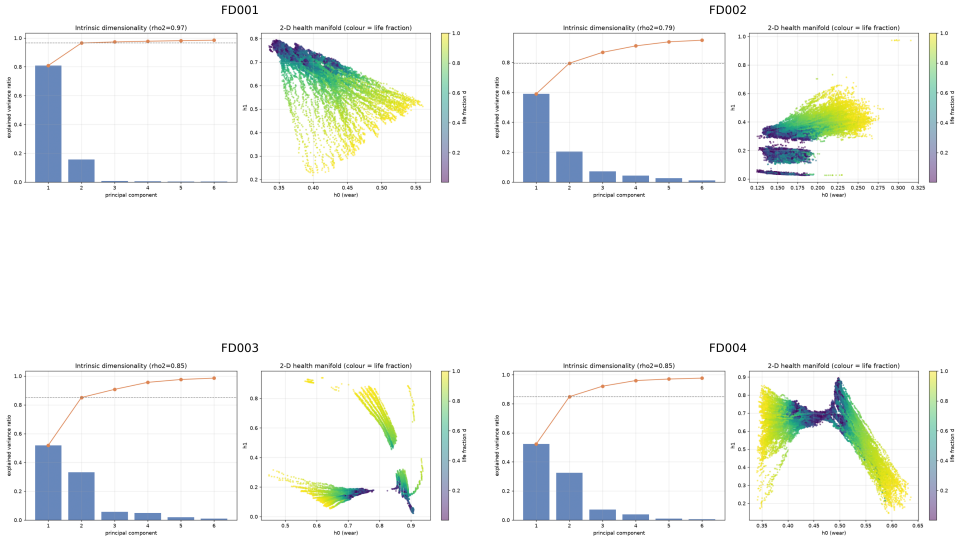


Figure 2: Intrinsic dimensionality and learned latent manifolds across C-MAPSS.

Learning Bounded Latent Degradation Dynamics

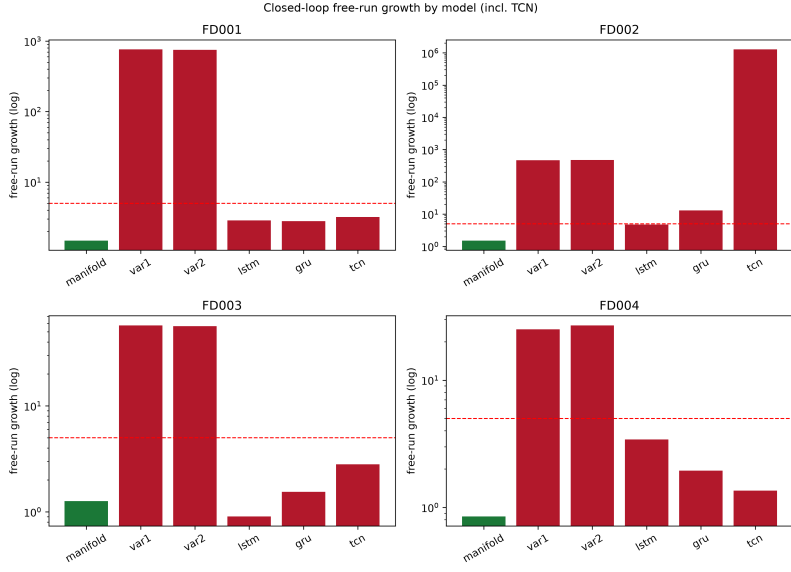


Figure 3: Free-run growth comparison across rollout models.

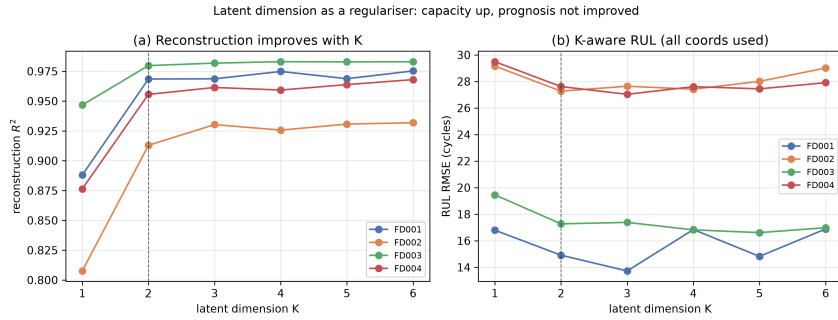


Figure 4: Latent dimension tradeoff between reconstruction and prognosis.

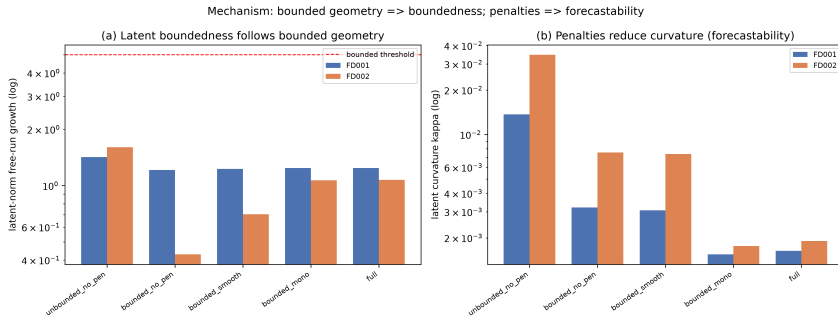


Figure 5: Mechanism analysis separating boundedness from forecastability.

C3_health_vs_rul

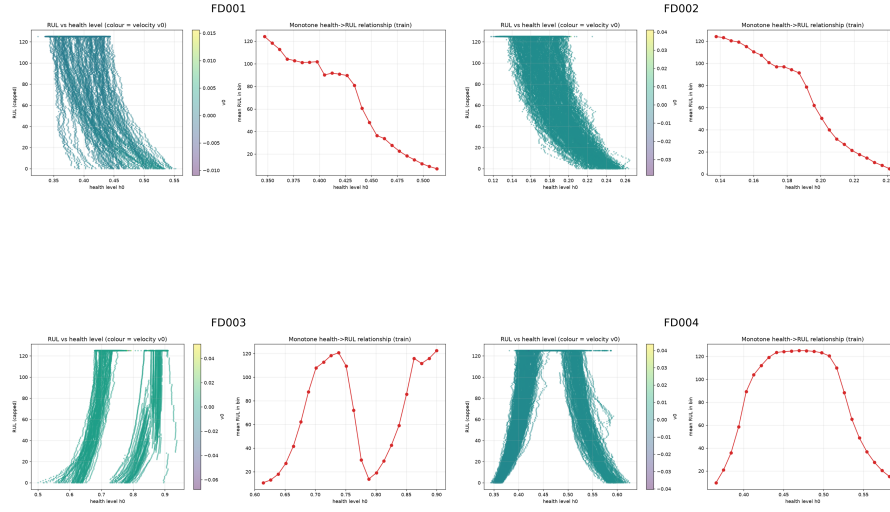


Figure 6: Learned latent health coordinate versus RUL.

A2_var_eigenvalues

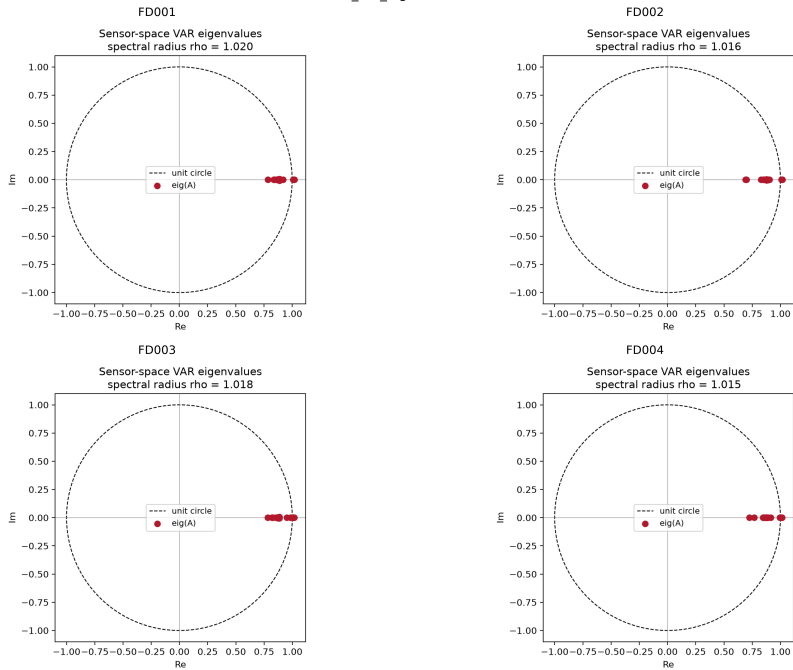


Figure 7: Sensor-space VAR eigenvalue spectra across FD001–FD004.

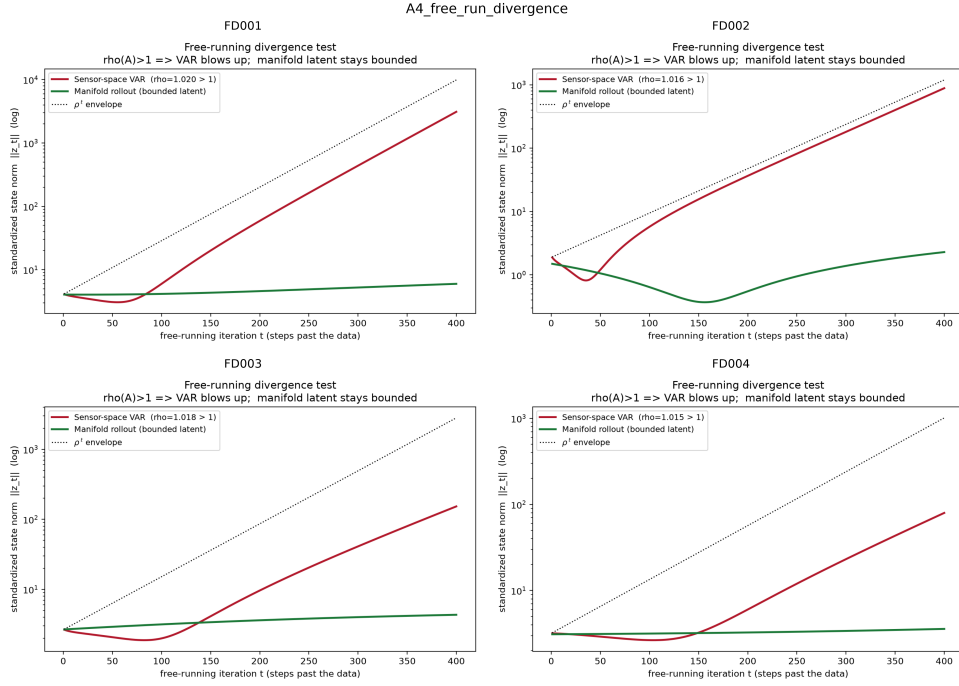


Figure 8: Free-run divergence trajectories for sensor-space VAR versus latent manifold rollout.

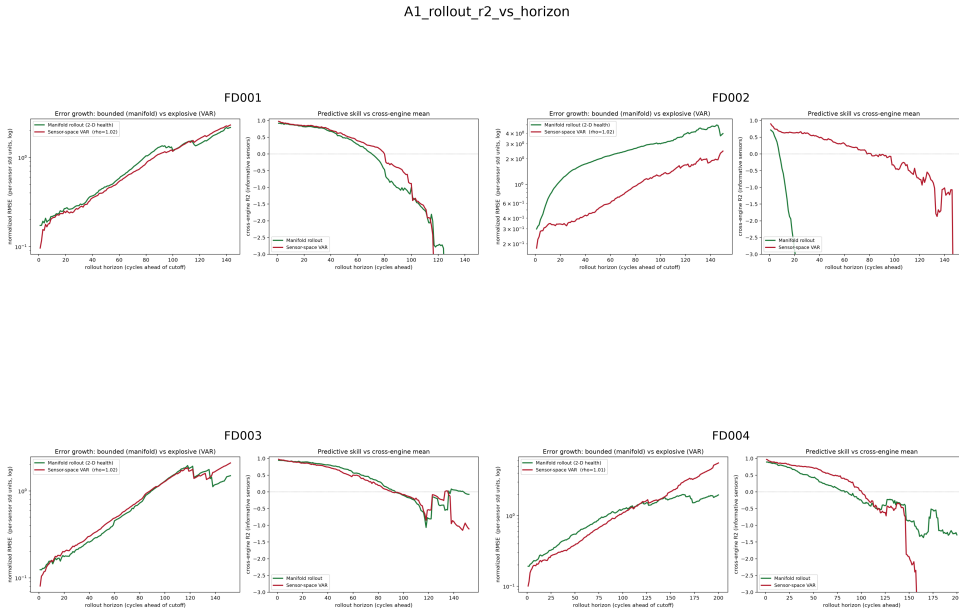


Figure 9: Rollout accuracy versus horizon.

A3_example_trajectories

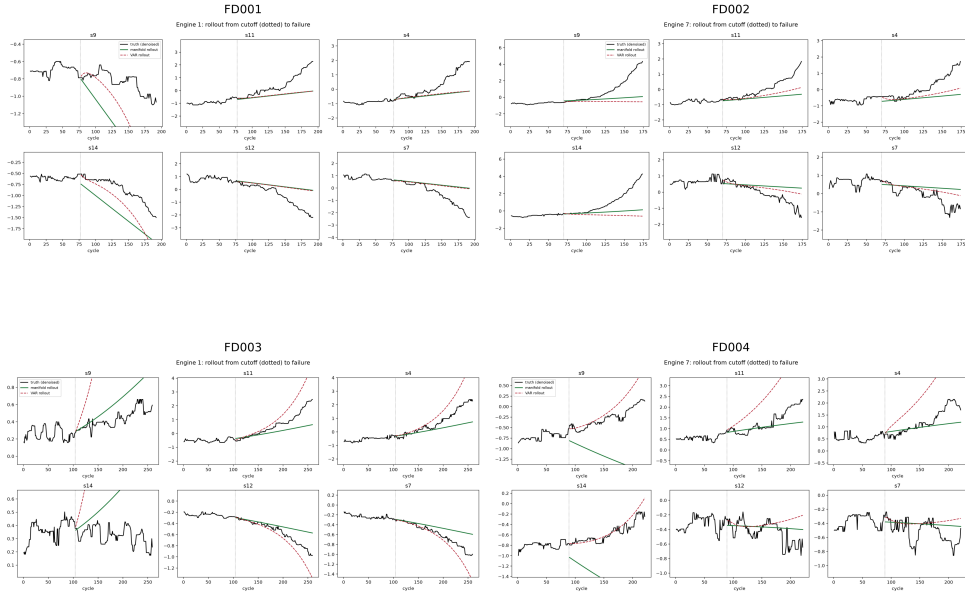


Figure 10: Representative sensor trajectories under true, manifold, and VAR rollouts.

B1_health_forecasts

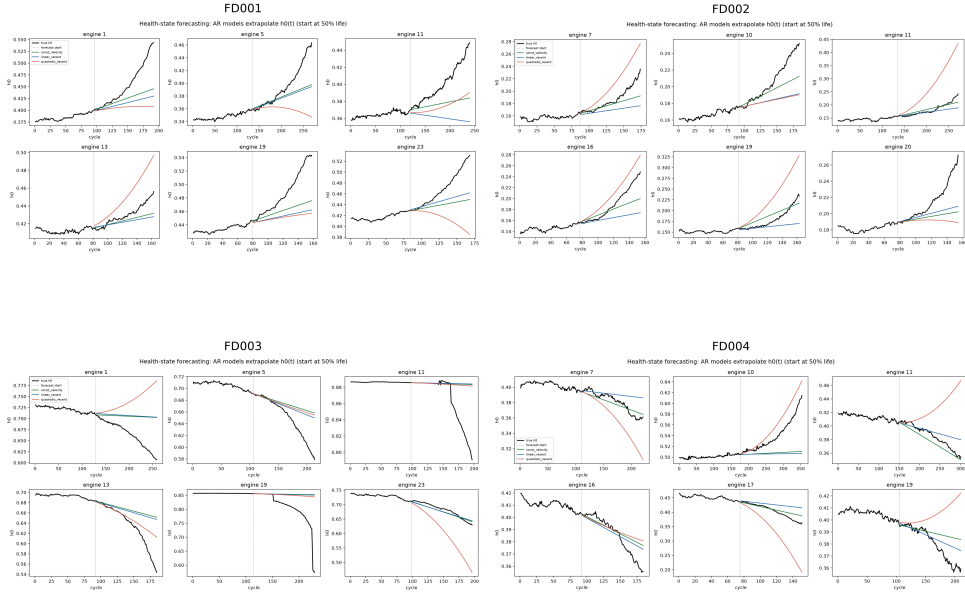


Figure 11: Example latent health forecasts.

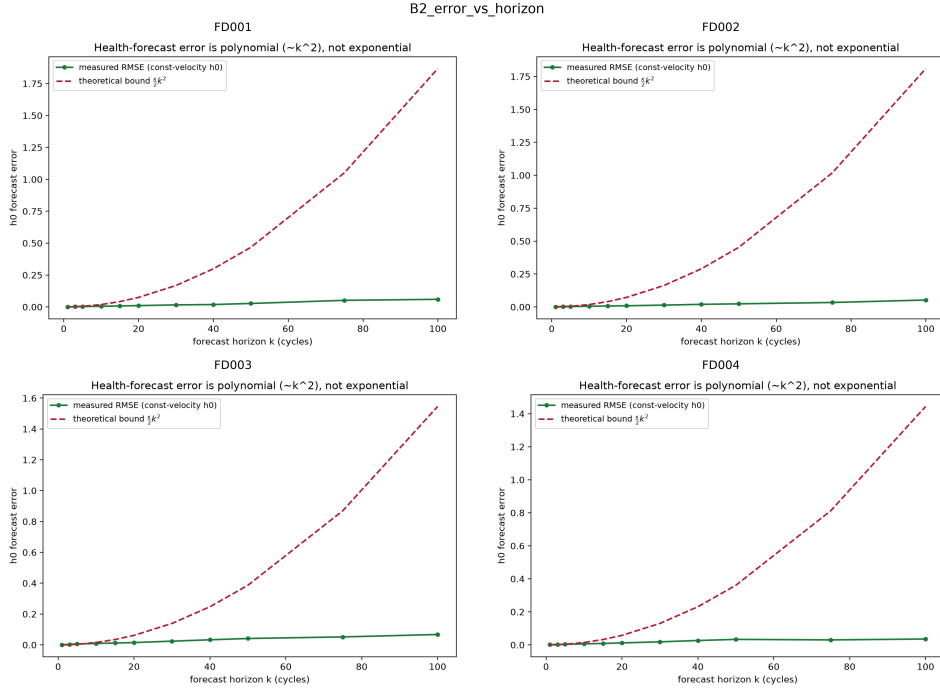


Figure 12: Forecast error growth versus horizon.

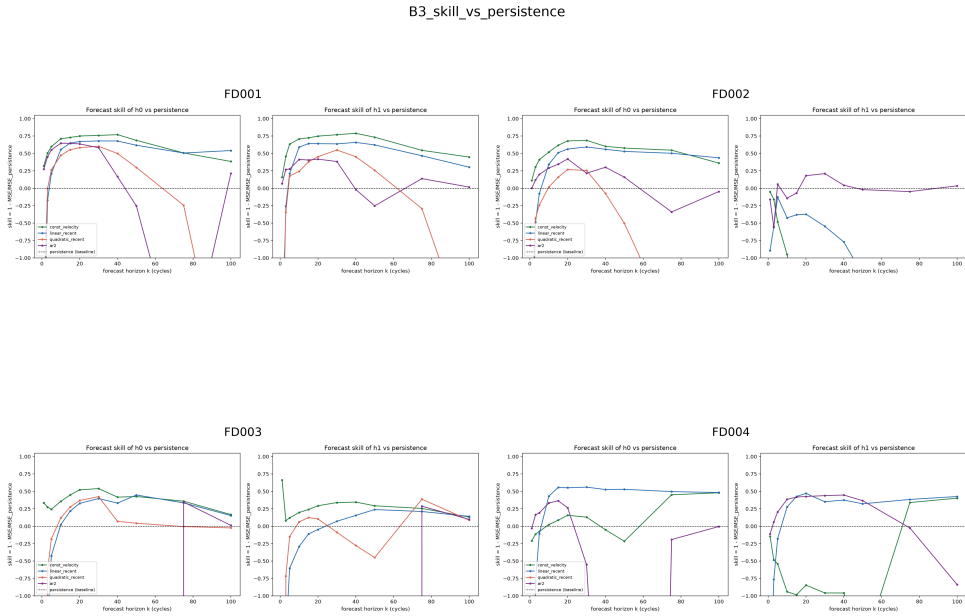


Figure 13: Forecast skill relative to persistence.

Learning Bounded Latent Degradation Dynamics

C1_rul_scatter

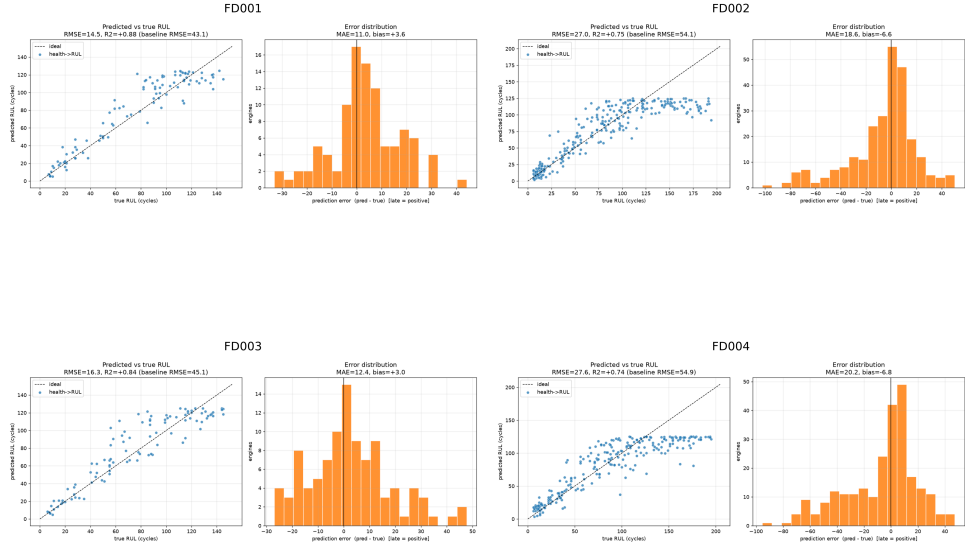


Figure 14: Predicted versus true RUL and error distributions.

C2_examples

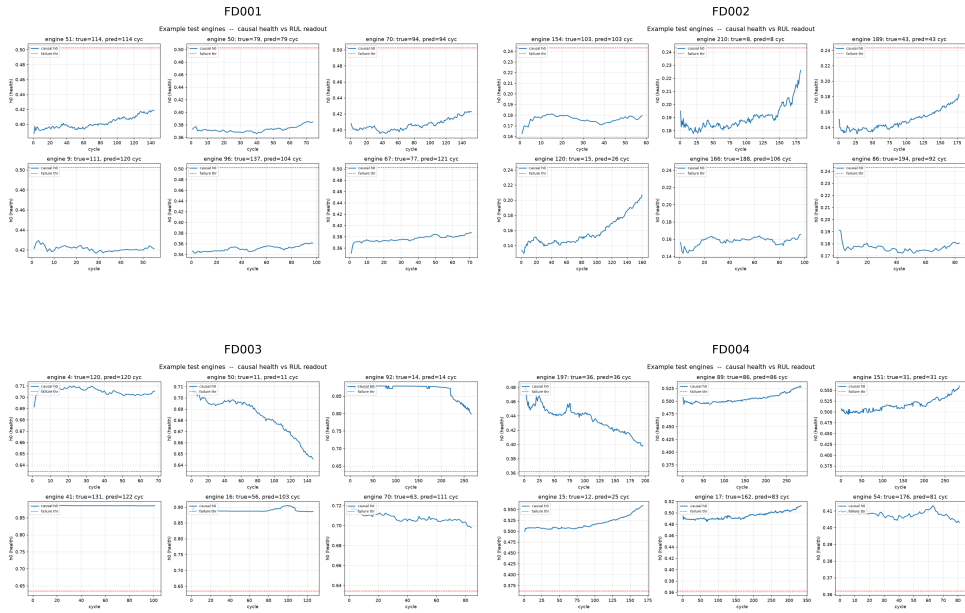


Figure 15: Representative test-engine RUL examples.

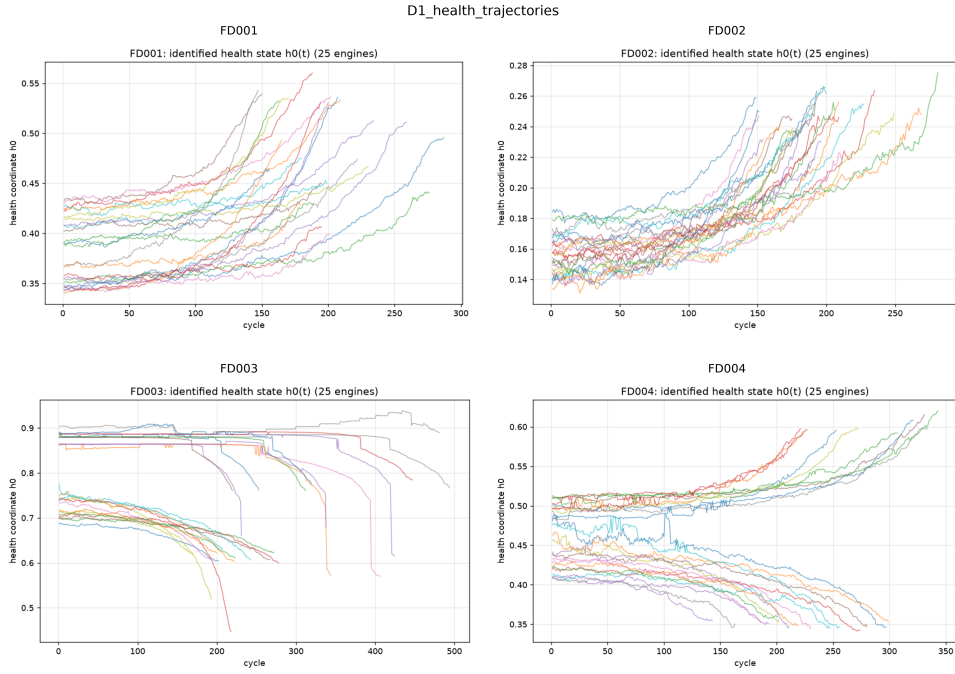


Figure 16: Primary health-coordinate trajectories across engines.

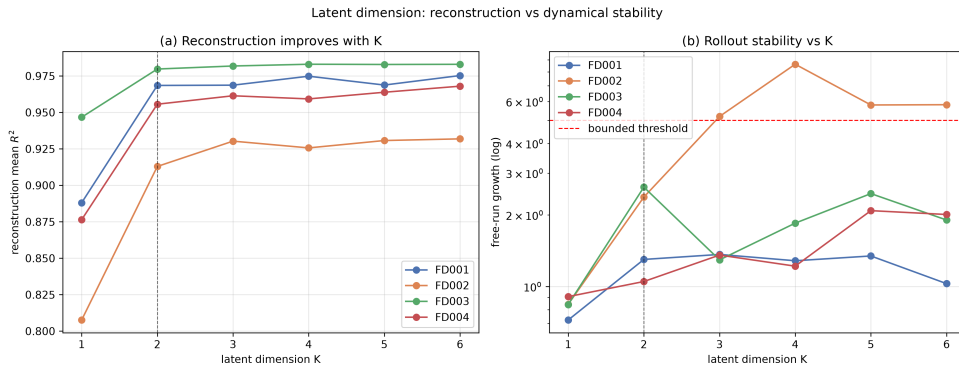


Figure 17: Expanded latent-dimension sweep showing reconstruction and stability axes.

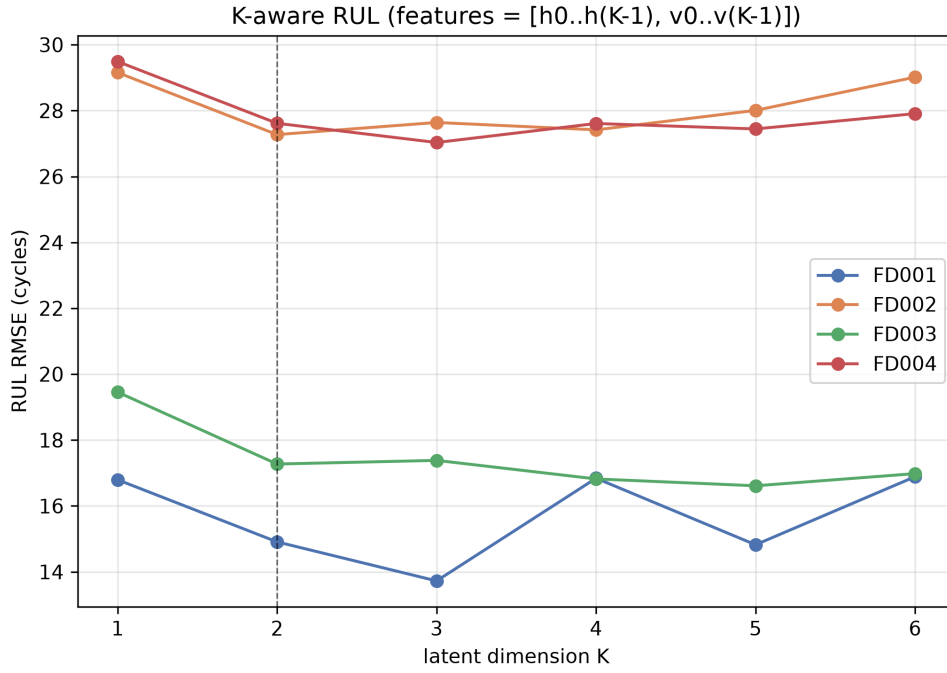


Figure 18: RUL RMSE versus latent dimension.

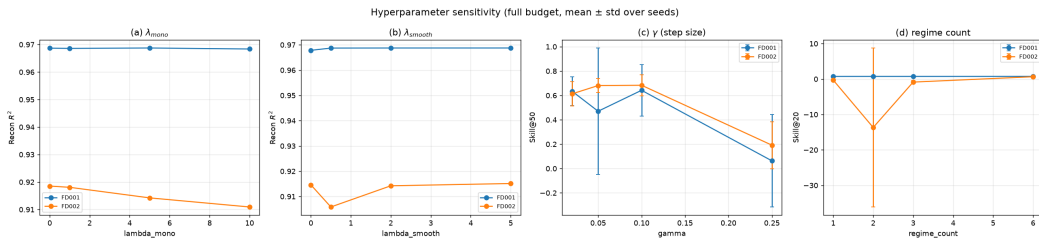


Figure 19: Full-budget hyperparameter sensitivity (mean $\pm$ std over three seeds). (a) monotonicity penalty, (b) smoothness penalty, (c) latent step size γ , (d) regime count.